INSTITUTO
DE INFORMÁTICA
UFRGS

Modeling Divergence Between Community-Perceived and Automatically Calculated MTG Commander Brackets

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CMP263 – Machine Learning
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DO RIO GRANDE DO SUL

Introduction.

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- **Commander**: *multiplayer* (4 players), 100-card deck (one copy of each), led by a legendary **commander**.
- **No matchmaking**: balance comes from a *pre-game* conversation.

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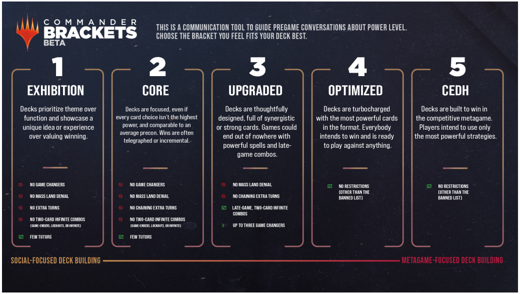

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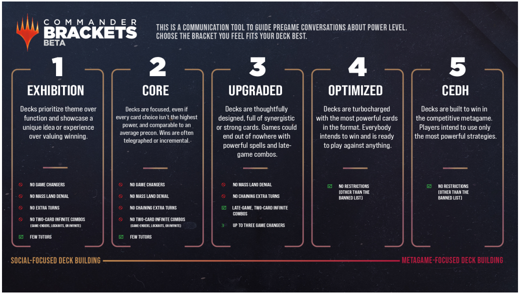

We focus on the middle — **brackets 2–4** — where almost all the public data lives and where the actual negotiation happens (1 and 5 are the extremes: thematic and cEDH).

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They frequently disagree.

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- Real problem for the Commander community.
- Case study of divergence between a human label and a heuristic label.

Methodology.

Roadmap.

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$$\boxed{\text{Collection}} \longrightarrow \boxed{\text{Preprocess}}$$

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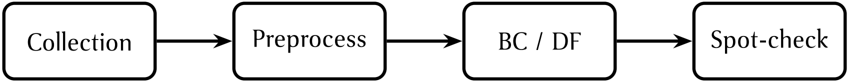

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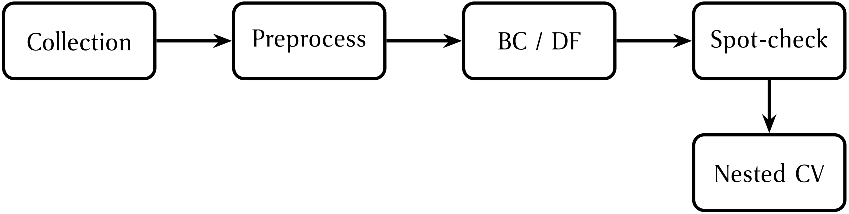

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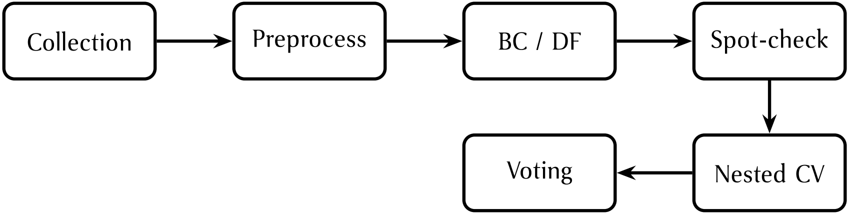

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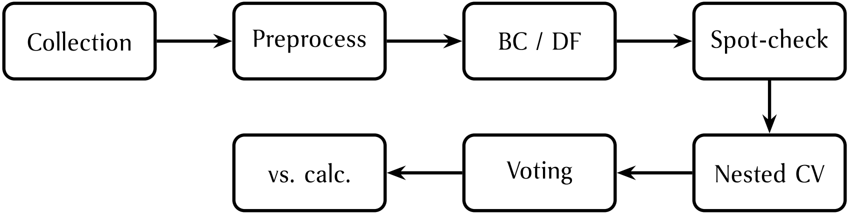
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BC / DF
Spot-check
Nested CV
Voting
vs. calc.

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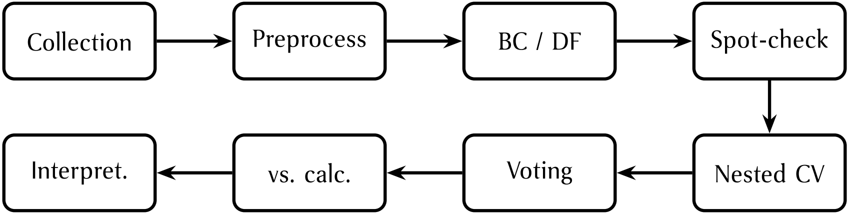
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Interpret.

Data.

Data collection.

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Archidekt

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14,421	decks collected
−1,471	duplicates
−815	$y_{\text{calc}} \in \{1, 5\}$
12,135	**modeling set**

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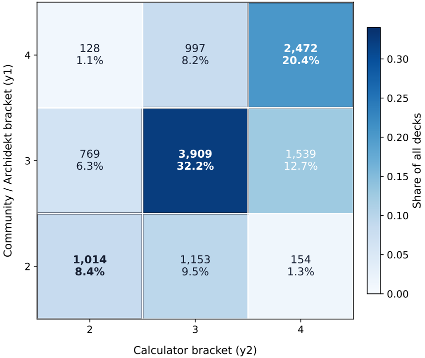

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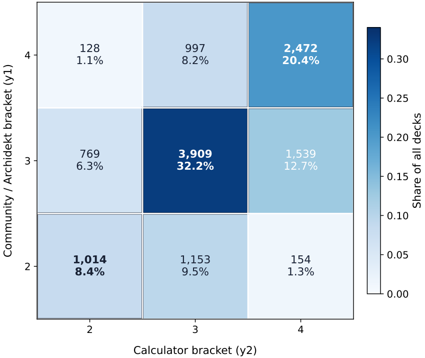

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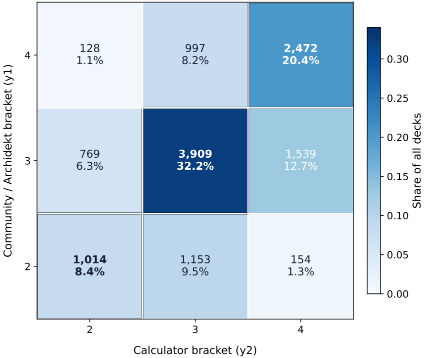

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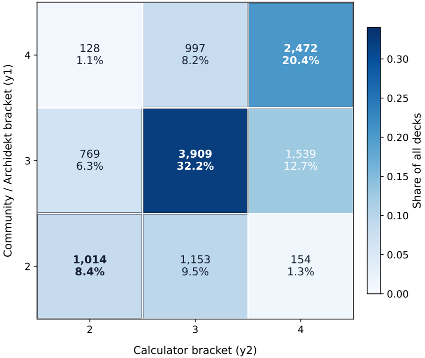

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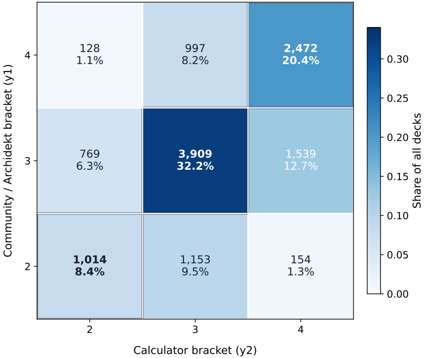

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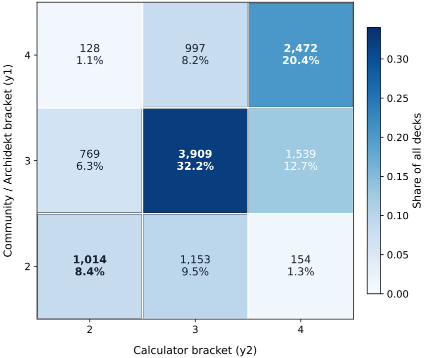

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  - **lower** in 15.6%

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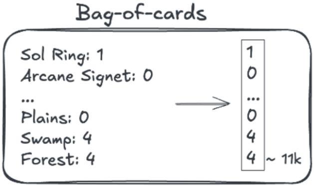

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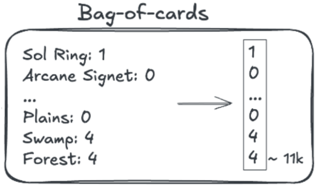

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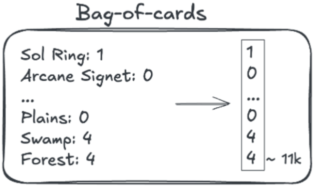

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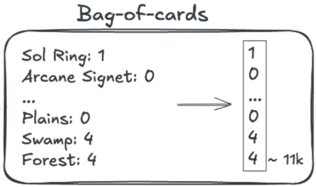

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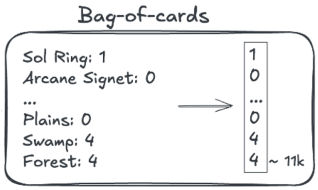

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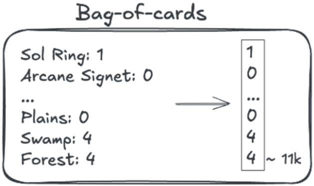

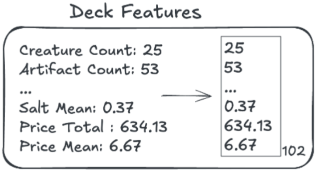

Results.

Spot-checking.

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Algorithm	DF	rank	BC	rank	in A_{union}
Gradient Boosting	0.6850	1	0.6342	1	✓
Random Forest	0.6686	2	0.5243	6	✓
Logistic Regression	0.6672	3	0.5927	2	✓
Linear SVC	0.6225	4	0.5459	4	✓
Decision Tree	0.6042	5	0.5445	5	✓
Naive Bayes	0.5792	6	0.5570	3	✓
KNN	0.5241	7	0.4541	7	—

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The 6 remaining algorithms $\times$ 2 representations $=$ **12 models** for nested CV.

Nested CV: DF beats BC for every algorithm.

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Algorithm	DF Macro-F1	DF Acc.	DF rank	BC Macro-F1	BC Acc.	BC rank
Gradient Boosting	0.6908 ± 0.009	0.698	1	0.6433 ± 0.012	0.644	1
Random Forest	0.6733 ± 0.008	0.703	2	0.6326 ± 0.012	0.653	2
Decision Tree	0.6727 ± 0.010	0.688	3	0.5475 ± 0.012	0.554	6
Logistic Regression	0.6707 ± 0.011	0.694	4	0.6236 ± 0.010	0.621	3
Linear SVC	0.6557 ± 0.009	0.658	5	0.6147 ± 0.012	0.628	4
Naive Bayes	0.5905 ± 0.009	0.587	6	0.5535 ± 0.009	0.564	5

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Best: **DF Gradient Boosting, F1 = 0.6908**.

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Case	Source	Exact	± 1	F1 calc	F1 arch
Raw labels	y_{arch}	60.9%	97.7%	0.5843	—
Max agreement	DF RF	69.3%	99.5%	0.6674	0.6733
Min agreement	BC DT	54.2%	96.6%	0.5328	0.5475
Best individual	DF GB	64.0%	98.1%	0.6323	0.6908
Best BC	BC GB	60.3%	97.3%	0.5984	0.6433
Best ensemble	Top-3 vote	66.7%	98.7%	0.6499	0.6941

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y_{calc} was **never used in training** — this is a purely descriptive comparison.

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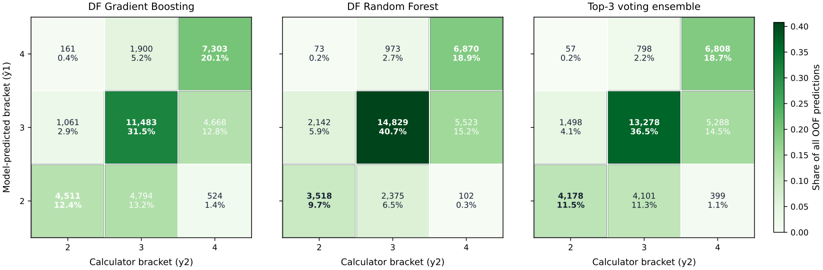

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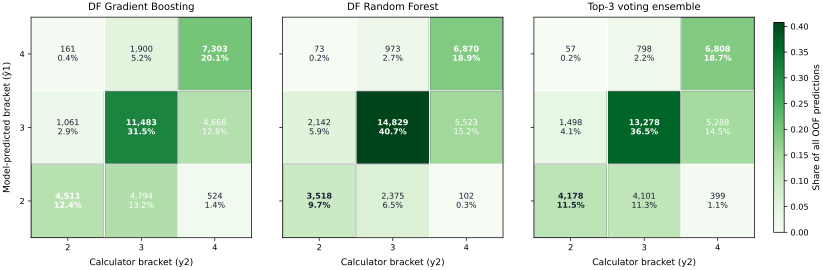

All three predictors sit **below** the calculator more often than above it — the same asymmetry seen in the raw community labels.

Interpretability.

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DF Gradient Boosting

permutation importance

Interpretability.

DF Gradient Boosting

permutation importance

Feature	PI
`game_changer_count`	**0.228**
`mass_land_denial_count`	0.029
`unique_atomic_combo_refs`	0.014
`tutor_count`	0.010
`cmc_1_count`	0.006

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high-lift cards in bracket 4

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Card	Lift
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Winter Moon	3.33
Chrome Mox	3.19
Imperial Seal	3.16
Grim Monolith	3.14

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Same signals, two views. DF sees the aggregates; BC sees the exemplar cards.

Discussion.

What we learned.

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What we learned.

- **DF $>$ BC everywhere.** Aggregate deck structure (102 features) outperforms card identity ($\approx$11k sparse dims) for every tuned algorithm — best F1: 0.69 vs 0.64.

- **Community $\neq$ calculator.** The best community predictor (DF GB) and the most calculator-aligned model (DF RF) are *different* models — both labels share Game Changers, combos, and tutors, but the community weights them differently.

What we learned.

- **DF > BC everywhere.** Aggregate deck structure (102 features) outperforms card identity (≈11k sparse dims) for every tuned algorithm — best F1: 0.69 vs 0.64.
- **Community ≠ calculator.** The best community predictor (DF GB) and the most calculator-aligned model (DF RF) are *different* models — both labels share Game Changers, combos, and tutors, but the community weights them differently.
- **Asymmetric disagreement.** When the two readings diverge, the calculator assigns a higher bracket in 23.5% of decks vs. lower in 15.6% — the community consistently rates itself below the calculator.

Demo.

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Reproducible pipeline — code, fixed seeds, saved folds and data snapshot (`uv run`).

Repository: `https://github.com/MarcoChit0/Machine-Learning-MTG`

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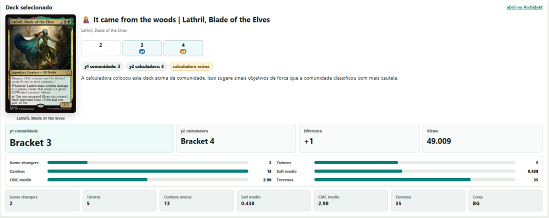

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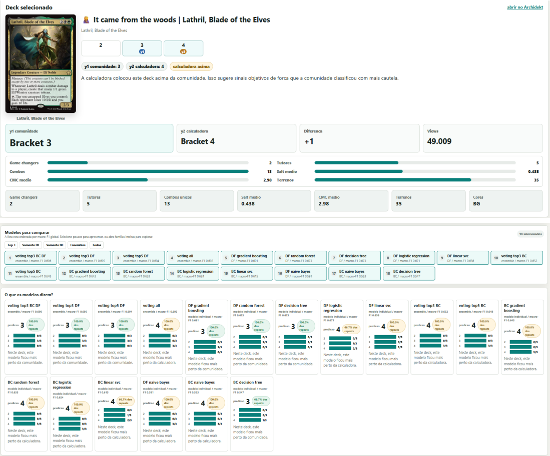

Thank you.

Questions?